

Van der Waals Equation of State for Real Gases

While developing Van der Waals equation of state for a real gas, the following simplifying assumptions are made. These assumptions are as follows:

- (i) Gas molecules have finite size (volume) and cannot be regarded as point masses.
- (ii) The gas molecules attract one another with a weak force of attraction that depends upon the distance between them.

Correction of pressure: Let us consider a gas molecule (labelled as A) well inside the vessel as shown in Fig. This molecule is influenced (attracted) by other molecules equivalently in all directions. As a result, net force acting on it is zero. The moment it strikes the wall of the vessel, it is pulled back by other molecules. Therefore, its velocity and hence momentum with which it strikes the wall would be less than the momentum with which it will strike in the absence of the intermolecular force of attraction. Therefore, the momentum transferred to the wall will be lower than in the case of an ideal gas (where intermolecular interactions are ignored). Therefore, molecular interactions cause a decrease in pressure. This decrease in pressure is known as *cohesive pressure*.



If we denote this drop by ΔP , cohesive pressure ΔP depends upon two factors.

- (i) the number of attracting molecules per unit volume (N/V)
- (ii) the number of molecules per unit volume (N/V) striking unit area of the walls of the containing vessel,

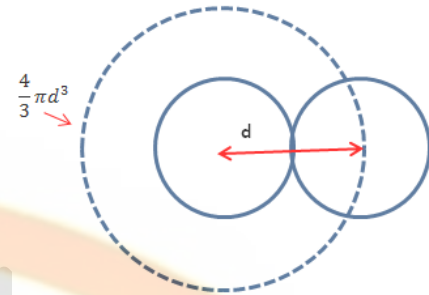
$$\Delta P \propto \left(\frac{N}{V}\right)^2$$

$$\Rightarrow \Delta P = \frac{a}{V^2}$$

Where a is called van der Waals constant and has the unit of $\text{N.m}^4\text{mol}^{-2}$.

Hence the real pressure = observed pressure + decrease in pressure = $P + \frac{a}{V^2}$

Correction of volume: If the finite size of molecules is taken into account, the volume available to them for random motion is somewhat less than the volume of the containing vessel. Consider two molecules which are in just in contact with each other. The minimum distance between their centers is equal to the molecular diameter d . The molecules cannot come closer than this distance. Let imagine a sphere of radius d around one molecule. The volume of this sphere is excluded for the center of the other molecule. This sphere is called the sphere of exclusion and has a volume equal to $\frac{4}{3}\pi d^3$. Thus for a pair of molecule, the unavailable volume is equal to the volume of sphere of exclusion. For a single molecule, the unavailable volume is half the volume of sphere of exclusion. Since there are N_A molecules in one mol of gas, the total unavailable volume per mol is. Thus the available volume per mol of a gas is $\frac{1}{2}[\frac{4}{3}\pi d^3]N_A$.



Thus the available volume per mole of a gas is $(V - \frac{1}{2}[\frac{4}{3}\pi d^3]N_A) = (V - b)$

Where $b = \frac{2}{3}\pi d^3 N_A = 4[\frac{4}{3}\pi (\frac{d}{2})^3]N_A = 4$ (Volume of a molecule)

The constant b has a unit of m^3/mol .

Therefore, with these corrections, the equation of state for an ideal gas changes to

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

This equation is called van der Waals equation of state for real gas.

Critical constants:

The van der Waals equation of state is

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

$$\Rightarrow PV - Pb + \frac{a}{V} - \frac{ab}{V^2} = RT$$

Multiplying by V^2 we have,

$$PV^3 - PbV^2 + aV - ab = V^2RT$$

Dividing by P and rearranging we have

$$V^3 - V^2 \left(b + \frac{RT}{P} \right) + V \frac{a}{P} - \frac{ab}{P} = 0 \quad \text{-----(1)}$$

This is a cubic equation in V and has three roots for a given value of P and T. At critical points all these roots will be equal since a gas has only one value of volume, pressure and temperature corresponding to it.

Let x be the value of the critical volume,

then $V = x = V - x = 0$

$$(V - x)^3 = 0$$

$$V^3 - 3xV^2 + 3x^2V - x^3 = 0 \quad \text{-----(2)}$$

Comparing the coefficients of V^3 , V^2 and V and the constant terms of (1) and (2)

$$3x = b + \frac{RT}{P} \quad \text{----- (3)}$$

$$3x^2 = \frac{a}{P} \quad \text{----- (4)}$$

And

$$x^3 = \frac{ab}{P} \quad \text{----- (5)}$$

Dividing (5) by (4) $x = 3b$

$$\therefore \text{Critical volume } V_c = 3b \quad \text{----- (6)}$$

Substituting the value of x in equation (4) we have $P = \frac{a}{27b^2} \quad \text{----- (7)}$

$$\text{Critical Pressure } P_c = \frac{a}{27b^2}$$

Putting the values of x and P in equation (3) we have

$$T = \frac{8a}{27Rb} \quad \text{----- (8)}$$

Thus the critical temperature, $T_c = \frac{8a}{27Rb}$

$$\therefore \frac{P_c V_c}{T_c} = \frac{3}{8} R$$

$$\Rightarrow R = \frac{8}{3} \frac{P_c V_c}{T_c}$$

The quantity $\frac{RT_c}{P_c V_c}$ is called critical coefficient of a gas.